From forces to fading motion

What makes an oscillator cross equilibrium while its motion shrinks?

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How does a released mass move while a spring restores it and a damper removes energy? Follow forces → differential equation → roots → real motion → initial data, then apply the chain to a new problem.

Prerequisites: Newton’ s law, time derivatives, the product rule and complex numbers. Here x' is velocity and x'' acceleration. We develop the differential-equation method when it becomes necessary.

$$mx'' = F_{\mathrm{s}} + F_{\mathrm{d}} = -kx - bx'$$

Choose $x = 0$ at equilibrium. The spring force opposes displacement; the viscous damper force opposes velocity. Both are modeling assumptions, not consequences of Newton’ s law.

Symbol	Meaning in this step
$F_{\mathrm{s}}, F_{\mathrm{d}}$	Spring and damping forces (N); functions of displacement and velocity.

$$mx'' + bx' + kx = 0$$

Move the two forces to the left. The zero on the right states that no external driving force acts.

Symbol	Meaning, units and role
$t;\ x(t)$	Time (s); signed displacement from equilibrium (m), the unknown function.
$x'(t);\ x''(t)$	Velocity (m/s); acceleration (m/s²), derivatives of the same $x(t)$.
$m > 0$	Mass (kg), fixed.
$b > 0$	Viscous damping coefficient (kg/s = N • s/m), fixed.
$k > 0$	Spring stiffness (N/m), fixed.

Model boundary and next question

Use one coordinate, a linear spring, a velocity-proportional damper and constant m, b, k. Large deformations, dry friction and forcing require a changed equation. Our equation constrains motion, but does not yet give $x(t)$.

Choose an exponential mode

Use the constant-coefficient structure before solving for rates

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An exponential retains its shape when differentiated. That makes it a useful trial for a linear equation with constant coefficients; it is a method choice, not a claim that the visible motion is a simple exponential.

$$x = Ce^{\lambda t}, \qquad x' = \lambda x, \qquad x'' = \lambda^2 x$$

Differentiation multiplies the trial by λ once for each derivative.

Symbol	Meaning in this step
C	Nonzero trial coefficient (m).
λ	Possibly complex rate (s^{-1}), to be determined; $i^2 = -1$.

$$(m\lambda^2 + b\lambda + k)Ce^{\lambda t} = 0$$

Substitute the trial and its derivatives. Since C is nonzero and the exponential never vanishes, the coefficient must be zero.

The exponential factor stays nonzero, so its coefficient must vanish. The next page solves that characteristic equation and explains why the roots describe decay and oscillation.

Find modes the equation preserves

Why an exponential trial turns derivatives into algebra

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The trial from page 2 requires $m\lambda^2 + b\lambda + k = 0$. Its roots determine which rates satisfy the motion equation. Separate the damping rate from the natural angular frequency.

$$\lambda^2 + 2\gamma\lambda + \omega_0^2 = 0$$

Divide by $m > 0$ and introduce two rates to separate damping from spring–mass motion.

Symbol	Meaning in this step
$\gamma = \dfrac{b}{2m}$	Decay rate (s^{-1}), fixed and positive.
$\omega_0 = \sqrt{\dfrac{k}{m}}$	Undamped angular frequency (rad/s), fixed. Radians are dimensionless.

$$(\lambda + \gamma)^2 = \gamma^2 - \omega_0^2$$

Complete the square. Under $0 < \gamma < \omega_0$, the right side is negative, so the roots form a complex conjugate pair.

$$\lambda = -\gamma \pm i\omega_\mathrm{d}, \qquad \omega_\mathrm{d} = \sqrt{\omega_0^2 - \gamma^2}$$

Write the square root of the negative number as ±i times a positive rate.

Symbol	Meaning in this step
ω_d	Damped angular frequency (rad/s); real and positive only in the underdamped regime.

What the roots tell us

The real part $-\gamma$ produces decay; the imaginary parts $\pm\omega_\mathrm{d}$ produce oscillation. We still need a real displacement and two constants that fit the initial data.

Convert complex modes to real motion

The product rule carries both the decay and the oscillation

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Euler’ s identity connects a complex exponential to cosine and sine. Because the equation is linear with real coefficients, the real and imaginary parts of a complex solution are real solutions. Their linear combination can match two independent initial data.

$$e^{(-\gamma + i\omega_d)t} = e^{-\gamma t}[\cos(\omega_d t) + i\sin(\omega_d t)]$$

Use $e^{i\theta} = \cos\theta + i\sin\theta$; $\theta = \omega_d t$ is dimensionless.

$$x(t) = e^{-\gamma t}[A\cos(\omega_d t) + B\sin(\omega_d t)]$$

Use the two real modes. They are independent when $\omega_d > 0$: their initial displacement–velocity pairs are $(1, -\gamma)$ and $(0, \omega_d)$.

Symbol	Meaning in this step
A, B	Real coefficients (m), determined by initial conditions rather than by m, b, k alone.

For differentiation only, call the trigonometric bracket $y(t)$, in metres: $x(t) = e^{-\gamma t}y(t)$. This helper represents the bracket, not the displacement.

$$y' = -A\omega_d\sin(\omega_d t) + B\omega_d\cos(\omega_d t)$$

Differentiate the bracket. The chain rule contributes ω_d to both trigonometric derivatives.

$$x' = -\gamma e^{-\gamma t}y + e^{-\gamma t}y' = e^{-\gamma t}(y' - \gamma y)$$

Apply the product rule to the exponential and the bracket; neither is constant.

Why the decay term matters

Differentiating only the trigonometric bracket misses $-\gamma A$ in the initial velocity. That omission changes B and gives a solution with the wrong launch condition.

We now have the general underdamped form and its velocity. The next page uses $x(0)$ and $x'(0)$ to select one motion from this family.

Let the initial data choose the solution

Work the illustrated case, then distinguish displacement from envelope

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The new givens x_0 (m) and v_0 (m/s) are the signed initial displacement and velocity. A and B are the real coefficients carried over from page 4.

$$x(0) = x_0 = A, \qquad x'(0) = v_0 = -\gamma A + \omega_{\mathrm{d}} B$$

Set $t = 0$ in the displacement and in both product-rule terms: the exponential and cosine become 1, and the sine becomes 0.

$$A = x_0, \qquad B = \frac{v_0 + \gamma x_0}{\omega_{\mathrm{d}}}$$

Rearrange the initial-velocity equation; division is allowed because $\omega_{\mathrm{d}} > 0$. Units of the numerator are m/s, so B has units m.

Illustrated baseline: $m = 1$ kg, $b = 0.4$ kg/s, $k = 4$ N/m, $x_0 = 0.10$ m and $v_0 = 0$. This is a new numerical illustration of the general model.

Quantity	Baseline calculation
γ; ω_0	0.200 s^{-1}; 2.00 rad/s, respectively.
ω_{d}	1.989974874 rad/s; 0.2 < 2 confirms underdamping.
A; B	0.100 m; 0.010050378 m, respectively.

$$R = \sqrt{A^2 + B^2}, \qquad |x(t)| \le R e^{-\gamma t}$$

The bracket is a scalar product of (A, B) with (cos θ, sin θ), a unit vector. Its absolute value cannot exceed $\sqrt{A^2 + B^2}$.

Symbol	Meaning in this step
R	Constant coefficient of the envelope (m); here $R \approx 0.100503782$ m.

The envelope is not the initial displacement

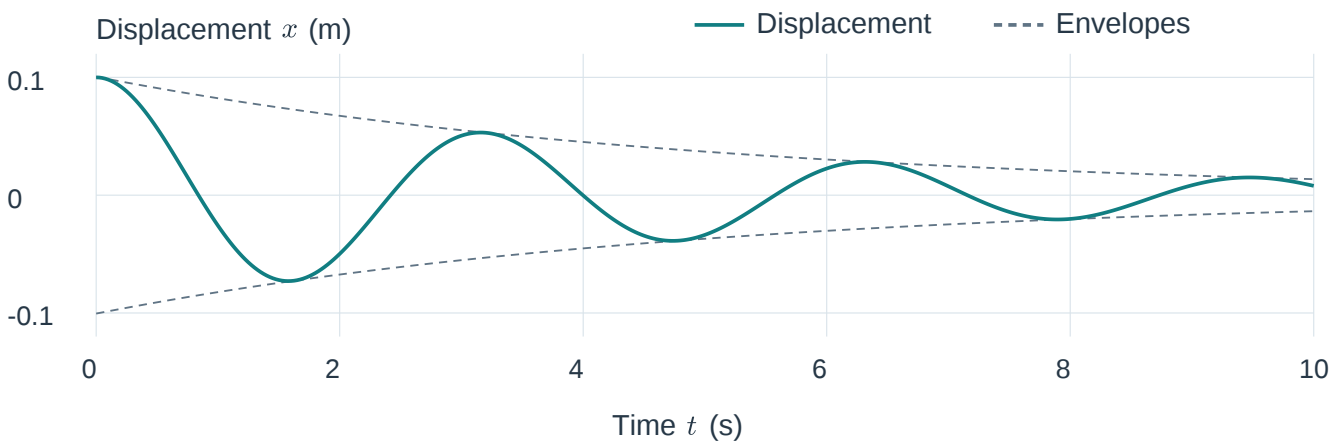
$A = x_0 = 0.10$ m, but $R \approx 0.100504$ m because B is nonzero. Release from rest does not require $B = 0$: the exponential also contributes to $x'(0)$. The next page shows this bound over the complete time response.

Read oscillation and decay in one graph

Use the initial-data solution to distinguish a motion from its bound

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The time-response chart uses the baseline from page 5. Its solid curve is displacement; the two dashed curves are $\pm Re^{-\gamma t}$. Read time in seconds and displacement in metres. The zero crossings show oscillation; the shrinking bounds show decay.



Envelopes: $\pm Re^{-\gamma t}$, $R = \sqrt{A^2 + B^2}$. Displacement still oscillates.

Original response and envelope plot; source slide 3. Same baseline as page 5.

The dashed curves are bounds, not a line connecting the turning points: the changing exponential shifts extrema away from cosine maxima. The envelope time $\tau = \frac{1}{\gamma}$ is 5 s: the envelope falls to e^{-1} of its initial value in 5 s. The oscillatory factor repeats after $T_\mathrm{d} = \frac{2\pi}{\omega_\mathrm{d}} \approx 3.15742$ s.

What this figure adds

The solid curve crosses equilibrium while the dashed bounds shrink. R depends on both launch conditions; the envelope bounds $x(t)$ rather than tracing it. Next we use the force law to account for the decreasing mechanical energy.

Connect the force law to energy loss

An exact power balance explains the decay without assuming a fixed sinusoidal amplitude

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The graph from page 6 shows the motion and its bounds. The force law from page 1 supplies an independent check: damping should remove energy whenever the mass moves.

$$E = \frac{1}{2}m(x')^2 + \frac{1}{2}kx^2$$

Mechanical energy is kinetic plus spring potential energy.

Symbol	Meaning in this step
E	Mechanical energy (J), a time-dependent quantity.

$$\frac{dE}{dt} = mx'x'' + kxx' = x'(mx'' + kx)$$

Apply the chain rule with m and k fixed, then factor out the velocity.

$$\frac{dE}{dt} = -b(x')^2 \leq 0$$

The equation of motion gives $mx'' + kx = -bx'$. Damping removes power at a nonnegative rate; kg/s × (m/s)² has units W.

A useful check

At release from rest, $\frac{dE}{dt} = 0$ at that instant, but energy decreases once the mass moves. The identity above is exact; it does not by itself make E(t) exactly proportional to $e^{-2\gamma t}$.

Original practice: select and predict a new motion

Full four-part question; worked part (a) starts below

Tutor-authored practice P1 • all conditions and parts

A free mass–spring–damper system has $m = 2.0$ kg, $b = 1.2$ kg/s and $k = 18$ N/m. At $t = 0$, $x_0 = 0.080$ m and $v_0 = -0.030$ m/s. The spring and damper remain linear and the coefficients remain constant.

Part	Task and lesson link
(a)	Classify the damping and calculate both characteristic roots. Use page 3.
(b)	Find A and B, then write the complete real solution $x(t)$. Use pages 4–5.
(c)	Calculate $x(1.5\,\mathrm{s})$ to three significant figures. Check it against the envelope bound. Use page 5.
(d)	Reset to the same m, k, x_0 and v_0, but double b to 2.4 kg/s. Check the regime and compare γ, ω_d, the envelope time τ (the reciprocal of γ) and the normalized envelope at 1.5 s. Recalculate R and explain whether equal x_0 means equal R. Use pages 3, 5–7.

Method choice

Use the underdamped solution only after checking $\gamma < \omega_0$. This is an initial-value problem with no driving force, so two launch conditions select A and B.

$$\gamma = \frac{1.2}{2(2.0)} = 0.300\ \mathrm{s^{-1}}, \qquad \omega_0 = \sqrt{\frac{18}{2.0}} = 3.00\ \mathrm{rad/s}$$

Part (a): substitute the new parameters into the definitions. Here 0.300 < 3.00, so the system is underdamped.

$$\omega_\mathrm{d} = \sqrt{3^2 - 0.3^2} = 2.984962311\ \mathrm{rad/s}$$

The negative discriminant produces two complex conjugate roots.

$$\lambda = -0.300 \pm 2.984962311i\ \mathrm{s^{-1}}$$

Both roots have a negative real part and a nonzero imaginary part: decaying oscillation.

Continue with worked parts (b)–(d) on pages 9–10. These answers use the practice data above, not the chart baseline.

Predict motion and check the launch data

Original P1(b)–(c) / Select the coefficients and check the envelope

Tutor-authored solutions P1(b)–(c)

$$B = \frac{-0.030 + (0.300)(0.080)}{2.984962311} \approx -0.00201008\ \text{m}$$

Part (b): $A = 0.080$ m from the displacement condition. The velocity relation gives B as above. The initial velocity −0.030 m/s includes the exponential derivative; it is not simply $\omega_{\text{d}} B$.

$$x(t) = e^{-0.300t}[0.080\cos(\omega_{\text{d}} t) - 0.00201008\sin(\omega_{\text{d}} t)]$$

Use $\omega_{\text{d}} = 2.984962311$ rad/s from page 8, metres for x and seconds for t. Displayed coefficients are rounded; calculations retain full precision. The launch velocity is −0.030 m/s.

Part (c): at $t = 1.5$ s, the phase is 4.477443467 rad and the exponential factor is 0.637628152. Substitute both into the solution, preserving the negative sine coefficient.

$$x(1.5\ \text{s}) = -0.010628207\ \text{m} \approx -0.0106\ \text{m}$$

The mass lies on the negative side of equilibrium. Its absolute displacement, 0.0106282 m, is below the envelope 0.0510264 m.

Part (d) continues on page 10. It resets to the same launch conditions and changes only the damping coefficient.

Change damping, then refit the launch data

Original P1(d) / Reset to the same mass, stiffness, displacement and velocity

Tutor-authored solution P1(d)

Part (d): change only b. The critical value is $b_c = 2\sqrt{mk} = 12$ kg/s, so $b = 2.4$ kg/s remains underdamped. Keeping the same launch conditions requires recalculating B and R.

Quantity	b = 1.2 kg/s	b = 2.4 kg/s
γ (s^{-1})	0.300	0.600
ω_d (rad/s)	2.984962	2.939388
τ (s)	3.33333	1.66667
B (m)	−0.00201008	0.00612372
R (m)	0.08002525	0.08023403
$e^{-\gamma \times 1.5}$	0.637628	0.406570
$Re^{-\gamma \times 1.5}$ (m)	0.05102635	0.03262072

The decay rate doubles and the envelope time halves, while ω_d decreases about 1.53%. The normalized envelope isolates this rate change. Equal x_0 does not make R equal: B depends on both changed γ and ω_d through the initial-velocity equation on page 5. R is also slightly larger than x_0 in both cases.

Do not compare displacement using the envelope alone

A faster-decaying envelope does not make $|x(t)|$ smaller at every instant, because phase and frequency also change. At 1.5 s the changed system has $x \approx -0.0120908$ m, even though its envelope is smaller.

Limits reveal when to change methods

Transfer the model; do not extend one regime beyond its condition

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Undamped limit: if b tends to zero with m and k fixed, γ tends to zero and ω_{d} tends to ω_0. The exponential disappears and the initial-condition solution becomes ordinary simple harmonic motion.

$$x(t) \longrightarrow x_0 \cos(\omega_0 t) + \frac{v_0}{\omega_0} \sin(\omega_0 t)$$

Take the limit of the complete solution, including B. The energy identity then gives $\frac{dE}{dt} = 0$.

Critical boundary: when γ tends to ω_0 from below, ω_{d} tends to zero. The apparent division by ω_{d} is resolved because $\frac{\sin(\omega_{\mathrm{d}} t)}{\omega_{\mathrm{d}}}$ tends to t; taking the limit of B alone would miss this cancellation.

$$x(t) \longrightarrow e^{-\omega_0 t}[x_0 + (v_0 + \omega_0 x_0)t]$$

Take the limit of the complete expression. At critical damping the repeated-root solution is no longer a sinusoid. For b greater than $2\sqrt{mk}$, use two real exponential modes instead.

Original transfer check

Double m while keeping b and k fixed. Must an originally underdamped system become overdamped? Use the definitions before answering.

$$\gamma_{\mathrm{new}} = \frac{\gamma}{2}, \qquad \omega_{0,\,\mathrm{new}} = \frac{\omega_0}{\sqrt{2}}, \qquad \frac{\gamma_{\mathrm{new}}}{\omega_{0,\,\mathrm{new}}} = \frac{1}{\sqrt{2}} \frac{\gamma}{\omega_0}$$

Answer: no. The ratio of decay rate to undamped angular frequency becomes smaller, so an underdamped system stays underdamped. This prediction assumes the same b and k remain valid when m changes.

Recognition cues: force balance establishes the model; constant coefficients invite exponential trials; complex conjugate roots invite decaying sine and cosine; initial displacement and velocity fix both coefficients. Check the regime before dividing by ω_{d}.

Useful next attempt: solve P1 again from the force law without looking at pages 8–10. Check the force units, the initial velocity and the envelope bound. Receiving this guide does not demonstrate mastery.